Uncertainty and Data Visualization

Exam-Ready Notes — Module 01, NPTEL: Statistical & Machine Learning Methods (Prof. Rajib Maity, IIT Kharagpur)

1. Definition

Uncertainty refers to a situation where information about a natural or real-world system is **incomplete, imprecise, or imperfect**. Statistical/ML models are based on assumptions and simplifications, so uncertainty is an **inherent part** of engineering analysis, planning, and design — decisions must be made even when future outcomes are not known with certainty.

2. Sources of Uncertainty

Source	Meaning	Reducible?
Inherent randomness	Real-world processes vary naturally over time/space (physical, chemical, biological, social factors)	**No** — unavoidable/irreducible
Model uncertainty	Models simplify real processes; assumptions may not fully capture actual behaviour	Reducible — better physical representation
Parameter uncertainty	Inaccurate parameter estimation during calibration / changes in system conditions	Reducible — better representation, realistic models
Data uncertainty	Data handling errors, measurement errors, inconsistent/non-homogeneous data	Reducible/avoidable — improve data quantity & quality
Operational uncertainty	Human errors during implementation/execution (manufacturing, construction, maintenance)	Minimizable — good workmanship, QC practices

3. Types of Uncertainty

Aleatory Uncertainty

- Originates from inherent natural variability and randomness
- **Cannot be eliminated** through additional information → **irreducible**

Epistemic Uncertainty

- Results from incomplete knowledge, data limitations, measurement inaccuracies
- **Can be reduced** through improved observations, data collection, understanding

	Aleatory	Epistemic
Origin	Natural randomness	Lack of knowledge/data
Reducibility	Irreducible	Reducible

4. Analysis of Uncertainty

Tasks involved:

- Quantification of the associated uncertainty
- Assessment of confidence level in model outputs
- Determination of contribution of input variables/parameters to output uncertainty
- Identification of most sensitive inputs/parameters

Components:

- **Qualitative** — identifies & classifies sources of uncertainty present in the system
- **Quantitative** — measures magnitude of uncertainty; evaluates impact on outputs using statistical/ML models
- **Uncertainty propagation** — how uncertainties in inputs/parameters propagate through the model to influence outputs (Input Uncertainty → Model → Output Uncertainty)
- **Communication** — conveying uncertainty to stakeholders/decision-makers

5. Measures of Uncertainty

Variance — describes how far observations are spread around the mean. Small variance → clustered near mean; large variance → widely dispersed.

Confidence Interval (CI) — a range within which the true value of a variable is expected to lie with a specified level of confidence.

Quartiles (divide ordered data into 4 equal parts):

- **Q1**: 25% of observations lie below this value
- **Q2 (Median)**: 50% of observations lie below this value
- **Q3**: 75% of observations lie below this value

$$IQR = Q3 - Q1$$

Represents the spread of the central 50% of the data.

6. Data Visualization

Data visualization is a **fundamental step** in statistical analysis and ML. Graphs give quick understanding of characteristics that are **easier to identify visually** than from numerical summaries alone.

Helps identify: Outliers • Missing values • Skewness • Trends and Seasonality • (to some extent) relationships between variables

Key point: Different datasets may produce similar statistical measures but exhibit completely different underlying patterns.

7. Representation of Data

Continuous time series f(t) are represented in discrete form for analysis/storage via two approaches:

Sample Data Representation

- Value = **instantaneous** value of the variable at a specific time instance
- Used when measurements are taken at regular time intervals
- Examples: streamflow, air temperature, wind speed, water level

$$X(i) = f(i\Delta t)$$

Pulse Data Representation

- Value = **total amount accumulated** over the interval — depends on the entire interval, not a single instant
- Example: **precipitation** (accumulated depths)

$$X(i) = \int_{(i-1)\Delta t}^{i\Delta t} f(t)\, dt$$

Average-rate form:

$$X(i) = (1/\Delta t) \int_{(i-1)\Delta t}^{i\Delta t} f(t)\, dt$$

8. Time Series Data

Time series = sequence of observations recorded in **chronological order** at regular or irregular time intervals.

- **Trend** — long-term increase or decrease in the data
- **Seasonality** — repeating pattern within a fixed time period
- **Cyclic variation** — long-term repeated fluctuations without a fixed period
- **Random variation/noise** — irregular fluctuations from uncertain/unexplained factors

9. Histogram

- Graphical representation used to display the **distribution** of numerical data
- Data grouped into a series of intervals (**bins**) of equal width
- Height of each bar = **frequency**
- Purpose: helps visualize descriptive statistics (mean, variance, skewness); gives a quick overview of data distribution

📌 Last-Minute Revision Box

- **Uncertainty** = incomplete/imprecise/imperfect information about a system
- **5 Sources:** Inherent randomness (irreducible) • Model • Parameter • Data • Operational (all reducible)
- **2 Types:** Aleatory (irreducible, natural) vs Epistemic (reducible, knowledge-gap)
- **Variance** = spread around mean | **CI** = range for true value at given confidence
- **Q1** = 25% below | **Q2** = median | **Q3** = 75% below
- **IQR = Q3 − Q1**
- **Sample:** $X(i) = f(i\Delta t)$ (instantaneous)
- **Pulse:** $X(i) = \int f(t)dt$ over interval (accumulated); rate form = $(1/\Delta t)\int f(t)dt$
- **Time series components:** Trend, Seasonality, Cyclic variation, Noise
- **Histogram** = frequency distribution using bins; bar height = frequency